Conditional VAE hyperparameters tuning procedure

Architectures We tested two architectures for the cVAE design (see Figure 1) with two configurations to integrate Y. In the first, denoted by 'full-cvae', we kept the encoding as is in jSAE. In the second, denoted by 'reduced-cvae', we introduced Y only at one layer at the end of the encoding part. Conditioning the decoder part on Y is done in the same way in both, combining the conditioning vector with a randomly generated latent vector, as input to the decoder. This allows us to test whether increasing the dependence of Z on Y pushes this network to ignore the conditioning signal at inference time as noted in [1]).

Training and validation procedure We ran training for 13 hyperparameters' configurations (see Table 1) with a noiseless training dataset. We ran each training for 300 epochs with a batch size of 300. We applied the learning rate scheduler $'on\_plateau'$ as we did in jSAE and used the estimation of $\mathcal{L}_{cVAE}$ on the validation set as the reference metric to track by the scheduler. When testing data normalization, we used the same robust Min-Max method we used with jSAE.

Table 1: Hyperparameters' configurations trained for cVAE

Experiment id	**KL weight** β	**Latent dimension** k	**Data normalization**	**Architecture**	**KL aggregation across the batch**	**Dataset size**
Id_0	1	30	TRUE	full-cvae	sum	7'000
Id_1	1	30	TRUE	reduced-cvae	sum	7'000
Id_2	1	30	FALSE	full-cvae	sum	7'000
Id_3	1	30	FALSE	reduced-cvae	sum	7'000
Id_4	1	60	TRUE	full-cvae	sum	7'000
Id_5	1	15	TRUE	full-cvae	sum	7'000
Id_6	10	60	TRUE	full-cvae	sum	7'000
Id_7	100	60	TRUE	full-cvae	sum	7'000
Id_8	1	90	TRUE	full-cvae	sum	7'000
Id_9	0.5	60	TRUE	full-cvae	sum	7'000
Id_10	0.25	60	TRUE	full-cvae	sum	7'000
Id_11	1	60	TRUE	full-cvae	average	7'000
Id_12	1	60	TRUE	full-cvae	sum	10'000

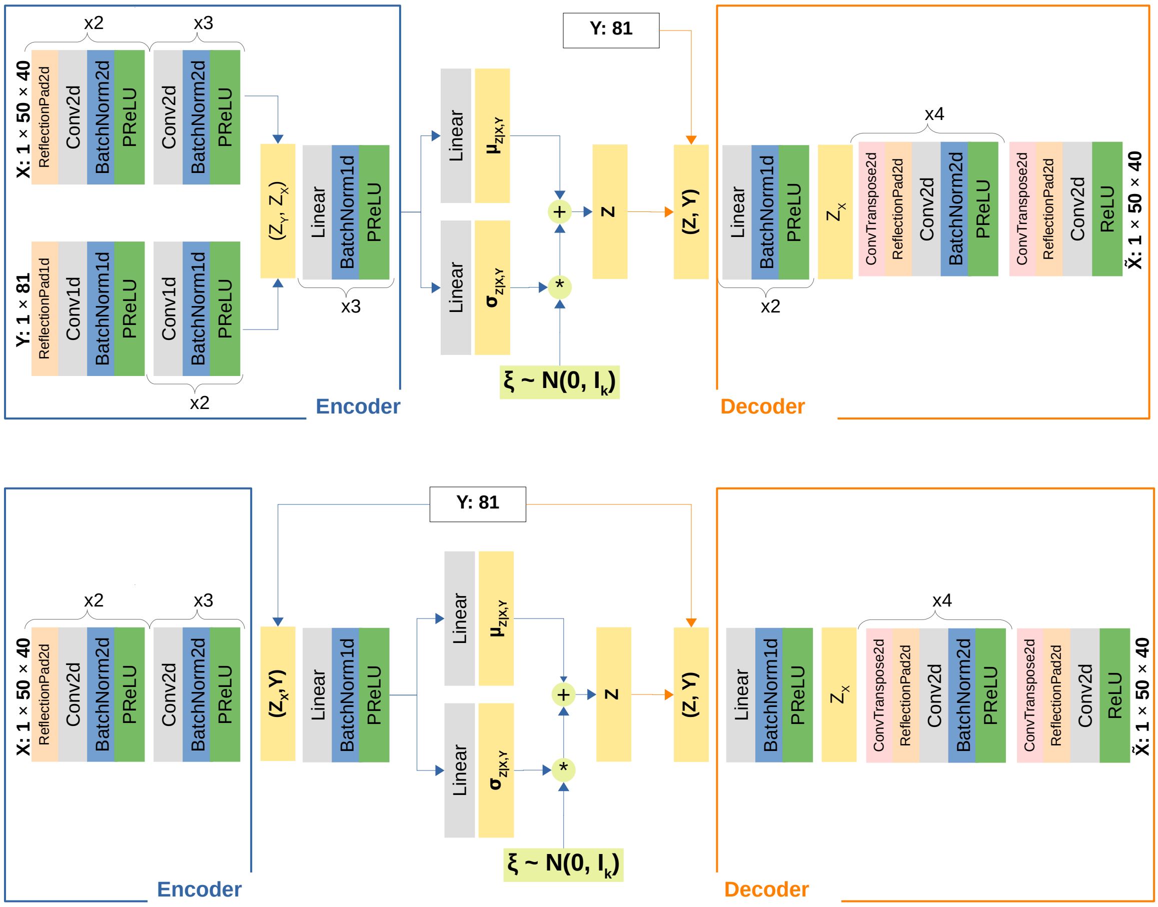

Figure 1: 'cVAE' architectures strting from our jSAE architecture: (upper) 'full-cvae', performs a full parallel encoding of Y; (lower) 'reduced-cvae', performs a reduced encoding of Y.

To compare the quality of probabilistic predictions from the trained configurations, we used the 60 test ground truths x^* we used to assess the sensitivity of FastABC. However, unlike what we did in the sensitivity assessment, for cVAE we contaminated y_{obs} with noise examples from the true noise distributions. Therefore, during this validation, we tested two noise scenarios, $\eta \sim \mathcal{N}(0, 0.5^2 I_{81})$ and $\eta \sim \mathcal{N}(0, 2.5^2 I_{81})$. Although this could be seen as leaking information about the unknown true noise into the hyperparameters, we treat this as giving cVAE more information than we authorize FastABC to see. Furthermore, we select the hyperparmeters based on the configuration that most minimizes, in both noise scenarios, the same inference metrics of $RMSE$, $ES_{1,2}$ and $VS_{0.5}$. Here, also, we intentionally inform cVAE about how its resulting D_{post} will be evaluated, while we never provide such information during the tuning of jSAE. Based on a posterior sample of size 500, we estimate each probabilistic metric 10 times by bootstrap with replacement. The summary statistics for those metrics are provided in Table 2. The configuration we retain corresponds to $\beta = 1$, $k = 60$, with robust Min-Max data normalization, using the summation to aggregate the batch of KL estimates and the 'full-cvae' architecture trained with 7'000 training examples (Experiment ID 4 in Table 2).

Overall, we found balancing the two parts of the loss to be challenging. Similar observations were already discussed in different works we encountered, notably [1] where they more extensively analyzed cVAE training challenges. In all configurations, the diversity of the inferred samples was very low, despite that the training did minimize the KL part. The decoder was able to change its predictions with different vector conditions y_{obs}. However, the samples for the same y_{obs} remained very close to each other. This indicates that the decoder network ignores the extra diversity of the realizations Z, sampled from $\mathcal{N}(0, I_k)$ at generation time, and only decodes the signal that was encoded about the relationship between Y and X. Although we tried to influence the balance with β, the latent space dimension k and the architecture, we were unable to increase the diversity of the posterior samples. We believe that the common recommendation to complement cVAE with an ad hoc sampler that learns to sample from the actual learned latent distributions $\mathrm{Q}_{\mathrm{Z|X,Y}}$, or their aggregate Q_{Z}, would be necessary to make these models work for inverse problems.

Table 2: Inference evaluation metrics $RMSE$, ES_1, ES_2 and $VS_{0.5}$ estimated with 60 test ground truths against a posterior samples generated by cVAE with different hyperparameters' configurations.

		$RMSE(x_{post}, x^*)$			$ES_1(D_{post}, x^*)$ $(\times 10^3)$			$ES_2(D_{post}, x^*)$ $(\times 10^3)$			$VS_{0.5}(D_{post}, x^*)$ $(\times 10^6)$		
	Exp._id	mean, std	median, IQR	[2.5q.- 97.5q.]	mean, std	median, IQR	[2.5q.- 97.5q.]	mean, std	median, IQR	[2.5q.- 97.5q.]	mean, std	median, IQR	[2.5q.- 97.5q.]
	Id_0	0.735, 0.231	0.685 [0.591 - 0.813]	[0.401 - 1.252]	1.153, 0.428	1.061 [0.863 - 1.249]	[0.551 - 2.119]	1.177, 0.8	0.935 [0.695 - 1.307]	[0.315 - 3.117]	0.8, 0.331	0.722 [0.572 - 0.939]	[0.364 - 1.615]
	Id_1	0.74, 0.234	0.687 [0.588 - 0.847]	[0.416 - 1.355]	1.163, 0.437	1.037 [0.899 - 1.308]	[0.603 - 2.37]	1.198, 0.814	0.924 [0.681 - 1.417]	[0.342 - 3.622]	0.8, 0.327	0.73 [0.587 - 0.951]	[0.377 - 1.593]
	Id_2	0.872, 0.22	0.823 [0.708 - 0.987]	[0.507 - 1.367]	1.415, 0.402	1.342 [1.123 - 1.619]	[0.778 - 2.233]	1.616, 0.852	1.353 [1.001 - 1.944]	[0.513 - 3.741]	0.964, 0.298	0.946 [0.75 - 1.087]	[0.523 - 1.692]
	Id_3	0.918, 0.217	0.865 [0.769 - 1.01]	[0.642 - 1.431]	1.479, 0.378	1.388 [1.252 - 1.582]	[0.995 - 2.461]	1.779, 0.902	1.499 [1.186 - 2.032]	[0.827 - 4.094]	0.963, 0.303	0.906 [0.778 - 1.075]	[0.49 - 1.771]
	Id_4	**0.727, 0.223**	**0.672 [0.552 - 0.832]**	**[0.433 - 1.228]**	**1.133, 0.409**	**1.018 [0.828 - 1.275]**	**[0.618 - 2.089]**	**1.146, 0.742**	**0.859 [0.601 - 1.366]**	**[0.36 - 2.948]**	**0.8, 0.325**	**0.707 [0.564 - 0.972]**	**[0.4 - 1.668]**
	Id_5	0.754, 0.224	0.713 [0.591 - 0.841]	[0.446 - 1.263]	1.171, 0.41	1.069 [0.881 - 1.336]	[0.617 - 2.126]	1.213, 0.771	0.999 [0.678 - 1.361]	[0.397 - 3.041]	0.805, 0.321	0.727 [0.586 - 0.908]	[0.395 - 1.564]
$\eta \sim \mathcal{N}(0, 0.5^2 I_{81})$	Id_6	0.722, 0.222	0.661 [0.569 - 0.852]	[0.426 - 1.212]	1.123, 0.411	0.997 [0.835 - 1.318]	[0.57 - 2.082]	1.131, 0.731	0.845 [0.651 - 1.428]	[0.349 - 2.837]	0.786, 0.323	0.686 [0.567 - 0.975]	[0.391 - 1.661]
	Id_7	0.726, 0.227	0.671 [0.568 - 0.842]	[0.422 - 1.226]	1.138, 0.423	1.035 [0.858 - 1.311]	[0.591 - 2.112]	1.15, 0.759	0.909 [0.66 - 1.377]	[0.347 - 2.948]	0.793, 0.327	0.704 [0.581 - 0.947]	[0.386 - 1.703]
	Id_8	0.73, 0.23	0.674 [0.565 - 0.81]	[0.416 - 1.262]	1.141, 0.428	1.023 [0.819 - 1.27]	[0.569 - 2.197]	1.16, 0.778	0.892 [0.637 - 1.273]	[0.34 - 3.182]	0.795, 0.324	0.725 [0.535 - 0.938]	[0.387 - 1.532]
	Id_9	1.006, 0.218	0.965 [0.85 - 1.141]	[0.66 - 1.72]	1.514, 0.396	1.444 [1.233 - 1.767]	[0.992 - 2.894]	2.095, 0.996	1.833 [1.389 - 2.576]	[0.847 - 6.009]	0.987, 0.331	0.906 [0.751 - 1.168]	[0.582 - 2.028]
	Id_10	0.726, 0.226	0.668 [0.568 - 0.838]	[0.432 - 1.219]	1.128, 0.415	1.023 [0.834 - 1.278]	[0.606 - 2.092]	1.145, 0.754	0.875 [0.649 - 1.382]	[0.356 - 2.915]	0.791, 0.325	0.703 [0.557 - 0.955]	[0.382 - 1.666]
	Id_11	0.722, 0.224	0.666 [0.562 - 0.825]	[0.432 - 1.222]	1.121, 0.415	1.012 [0.828 - 1.268]	[0.592 - 2.091]	1.132, 0.749	0.882 [0.637 - 1.334]	[0.35 - 2.867]	0.785, 0.324	0.699 [0.575 - 0.956]	[0.392 - 1.654]
	Id_12	0.726, 0.219	0.673 [0.562 - 0.856]	[0.425 - 1.247]	1.148, 0.412	1.02 [0.849 - 1.374]	[0.58 - 2.185]	1.142, 0.713	0.884 [0.635 - 1.472]	[0.36 - 3.107]	0.789, 0.308	0.713 [0.577 - 0.973]	[0.372 - 1.541]
	Id_0	1.151, 0.259	1.107 [0.964 - 1.318]	[0.793 - 1.658]	1.786, 0.451	1.729 [1.45 - 2.078]	[1.217 - 2.797]	2.77, 1.302	2.417 [1.863 - 3.438]	[1.262 - 5.415]	1.274, 0.333	1.196 [1.025 - 1.451]	[0.837 - 1.929]
	Id_1	1.181, 0.268	1.144 [0.961 - 1.368]	[0.74 - 1.902]	1.828, 0.461	1.712 [1.496 - 2.13]	[1.101 - 3.068]	2.912, 1.344	2.623 [1.865 - 3.714]	[1.09 - 6.909]	1.268, 0.308	1.236 [1.031 - 1.496]	[0.789 - 1.925]
	Id_2	1.494, 0.312	1.439 [1.243 - 1.716]	[1.036 - 2.167]	2.341, 0.456	2.29 [1.998 - 2.614]	[1.646 - 3.289]	4.657, 2.056	4.142 [3.099 - 5.88]	[2.146 - 9.388]	1.695, 0.395	1.676 [1.401 - 1.866]	[1.149 - 2.797]
	Id_3	1.711, 0.312	1.693 [1.477 - 1.888]	[1.206 - 2.499]	2.783, 0.522	2.775 [2.416 - 3.134]	[1.968 - 4.11]	6.05, 2.257	5.748 [4.386 - 7.103]	[2.908 - 12.492]	1.763, 0.416	1.658 [1.496 - 2.007]	[1.161 - 2.714]
	Id_4	**1.005, 0.207**	**0.971 [0.844 - 1.103]**	**[0.693 - 1.472]**	**1.542, 0.361**	**1.466 [1.306 - 1.705]**	**[1.015 - 2.367]**	**2.09, 0.891**	**1.869 [1.435 - 2.377]**	**[0.956 - 4.332]**	**1.09, 0.249**	**1.041 [0.924 - 1.221]**	**[0.675 - 1.672]**
	Id_5	1.288, 0.377	1.176 [0.998 - 1.499]	[0.856 - 2.506]	1.983, 0.635	1.875 [1.545 - 2.195]	[1.309 - 4.368]	3.555, 2.382	2.713 [2.009 - 4.515]	[1.463 - 12.445]	1.456, 0.545	1.284 [1.1 - 1.702]	[0.912 - 3.347]
$\eta \sim \mathcal{N}(0, 2.5^2 I_{81})$	Id_6	1.078, 0.238	1.017 [0.913 - 1.191]	[0.751 - 1.656]	1.644, 0.396	1.536 [1.404 - 1.857]	[1.06 - 2.628]	2.406, 1.125	2.055 [1.661 - 2.755]	[1.153 - 5.439]	1.172, 0.291	1.107 [0.987 - 1.28]	[0.681 - 1.849]
	Id_7	1.157, 0.288	1.112 [0.933 - 1.313]	[0.758 - 1.838]	1.768, 0.45	1.687 [1.442 - 1.968]	[1.142 - 2.891]	2.81, 1.459	2.476 [1.721 - 3.415]	[1.156 - 6.612]	1.28, 0.373	1.207 [1.016 - 1.455]	[0.757 - 2.359]
	Id_8	1.089, 0.246	1.043 [0.936 - 1.205]	[0.69 - 1.705]	1.678, 0.426	1.617 [1.453 - 1.824]	[1.035 - 2.743]	2.48, 1.174	2.149 [1.755 - 2.884]	[0.943 - 5.747]	1.184, 0.298	1.148 [1.059 - 1.286]	[0.675 - 1.946]
	Id_9	1.56, 0.289	1.552 [1.363 - 1.71]	[0.965 - 2.258]	2.388, 0.487	2.361 [2.091 - 2.644]	[1.49 - 3.623]	5.007, 1.92	4.829 [3.699 - 5.805]	[1.822 - 10.121]	1.738, 0.446	1.597 [1.435 - 1.999]	[1.117 - 2.833]
	Id_10	1.072, 0.231	1.024 [0.895 - 1.196]	[0.753 - 1.629]	1.649, 0.392	1.543 [1.379 - 1.851]	[1.108 - 2.669]	2.384, 1.067	2.097 [1.612 - 2.794]	[1.145 - 5.29]	1.178, 0.3	1.132 [1.008 - 1.321]	[0.725 - 1.802]
	Id_11	1.095, 0.243	1.045 [0.919 - 1.187]	[0.772 - 1.725]	1.667, 0.4	1.595 [1.416 - 1.822]	[1.094 - 2.717]	2.484, 1.168	2.182 [1.699 - 2.727]	[1.201 - 5.802]	1.193, 0.285	1.148 [1.024 - 1.314]	[0.714 - 1.949]
	Id_12	1.187, 0.281	1.129 [1.011 - 1.312]	[0.711 - 1.899]	1.842, 0.481	1.763 [1.494 - 2.056]	[1.07 - 3.184]	2.964, 1.53	2.532 [2.038 - 3.42]	[0.997 - 7.205]	1.32, 0.352	1.265 [1.099 - 1.468]	[0.831 - 2.29]

Bibliography

[1] C. Beckham, *Generalisation with Generative Models*. PhD thesis, Ecole Polytechnique, Montreal (Canada), 2024.